Neural Networks in Financial Prediction

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Abstract

This project studies cross-asset return for liquid U.S. equities. The prediction target is the five-class 1-horizon return label: $\{-2, -1, 0, 1, 2\}$. The model filtration is

$$\text{Logistic} \to \text{MLP} \to \text{GCN} \to \text{Multi-relatinal GCN}.$$

The experiment uses CRSP daily data from January 1, 2000 through December 31, 2025. The goal is to evaluate whether nonlinearity, graph convolution, higher-order structure of relations, and signal consistency improve generalization.

Contents

1 Ingredients

1.1 Liquid assets

Dataset is CRSP daily equity from January 1, 2000 through December 31, 2025. At each trading day $t \in \mathcal{T} = [\text{2000-01-01, 2025-12-31}]$, the investable asset universe $\mathcal{A}_t$ is formed from the top 500 U.S. equities by trailing dollar_volume of 21 days, i.e.

$$\mathcal{A}_t = \underset{i_1 \geq \ldots \geq i_{500}}{\arg\max} \left\{ \text{lagged}^{(21)}\text{\_dollar\_volume}_{i,t} \right\}.$$

Accoring to the following section [Raw data], we have

$$\text{dollar\_volume}_{i,t} = \left| \text{prc}_{i,t} \right| \cdot \text{vol}_{i,t},$$

$$\text{lagged}^{(21)}\text{\_dollar\_volume}_{i,t} = \frac{1}{21} \sum_{s \in W_{i,t}} \text{dollar\_volume}_{i,s}, \quad W_{i,t} = \{t-21, ..., t-1\}.$$

1.2 Target

For $\alpha \in \{20, 40, 60, 80\}$, define

$$q_{\alpha,t+1} = \text{Quantile}_\alpha \left(\{ r_{j,t+1} : j \in \mathcal{A}_t \} \right).$$

The five-class label is

$$y_{i,t} = \begin{cases} -2, & r_{i,t+1} \leq q_{20,t+1}, \\ -1, & q_{20,t+1} < r_{i,t+1} \leq q_{40,t+1}, \\ 0, & q_{40,t+1} < r_{i,t+1} \leq q_{60,t+1}, \\ 1, & q_{60,t+1} < r_{i,t+1} \leq q_{80,t+1}, \\ 2, & r_{i,t+1} > q_{80,t+1}. \end{cases}$$

1.3 Raw data

Raw data

$$\rho_{i,t} = \begin{pmatrix} \text{siccd} \\ \text{prc} \\ \text{ret} \\ \text{retx} \\ \text{vol} \\ \text{mktcap} \\ \text{open} \\ \text{high} \\ \text{low} \\ \text{close} \\ \text{delist\_flag} \end{pmatrix},$$

where $(i,t) = (\text{permno, date}) \in \mathcal{A}_t \times \mathcal{T}$ is the unique identifier.

1.4 Cellular structure

The 0-cells are assets

$$K_t^{(0)} = \{i \in \mathcal{A}_t\}.$$

There are five types of 1-cell: correlation, lead-lag, sector, supply-chain, market-exposure.

1.4.1 Correlation 1-cell

The 120-day rolling return correlation is

$$\rho_{ij,t}^{120} = \mathrm{Corr}\left(r_{i,t-120:t-1}, r_{j,t-120:t-1}\right).$$

Let $N_k(i,t)$ be the assets with the largest k values of $|\rho_{ij,t}^{120}|$.
The correlation adjacency is

$$A_{ij,t}^{\text{corr}} = \begin{cases} \mathbb{1}\{j \in N_k(i,t) \text{ or } i \in N_k(j,t)\}, & \rho_{ij,t} > 0 \\ -\mathbb{1}\{j \in N_k(i,t) \text{ or } i \in N_k(j,t)\}, & \rho_{ij,t} < 0 \end{cases}.$$

1.4.2 Lead-lag 1-cell

The 120-day 5-lagged correlation is

$$\lambda_{i \to j,t}^{120,5} = \mathrm{Corr}\left(r_{i,t-124:t-5}, r_{j,t-120:t-1}\right).$$

Let $N_k^{\text{lead}}(i,t)$ be the assets j with the largest k values of $|\lambda_{i \to j,t}^{120,5}|$.
The lead-lag adjacency is

$$A_{ij,t}^{\text{lead}} = \mathbb{1}\{j \in N_k^{\text{lead}}(i,t)\}.$$

1.4.3 Sector-industry 1-cell

The sector adjacency is

$$A_{ij}^{\text{sector}} = \mathbb{1}\{\text{sector}_i = \text{sector}_j\}.$$

$$A_{ij}^{\text{industry}} = \mathbb{1}\{\text{industry}_i = \text{industry}_j\}.$$

1.4.4 Supply-chain 1-cell

The supply-chain adjacency is

$$A_{i \to j,t}^{\text{sup}} = \mathbb{1}\{i \text{ supplies } j \text{ as of } t\}.$$

1.4.5 Market-exposure 1-cell

Define market-exposure similarity by

$$\phi_{ij,t} = \frac{\beta_{i,t}^\top \beta_{j,t}}{\|\beta_{i,t}\| \, \|\beta_{j,t}\|},$$

where β is the market exposure. Let $N_k^{\text{beta}}(i,t)$ be the assets j with the largest k values of $\phi_{ij,t}$. The market-exposure adjacency is

$$A_{ij,t}^{\text{beta}} = \mathbb{1}\{j \in N_k^{\text{beta}}(i,t) \text{ or } i \in N_k^{\text{beta}}(j,t)\}.$$

1.4.6 Comments on 1-cell

Random 1-cell versus Configured 1-cell For each of the five constructed 1-cell, we will use the same number random 1-cell to detect whether our inductive bias capture signals.

Top-k versus τ-Threshold Sensitivity For every top-k 1-cell construction, we evaluate

$$k \in \{3, 5, 10, 20\}.$$

Alternatively, we can construct τ-threshold 1-cell. We evaluate

$$\tau \in \{0.2, 0.4, 0.6, 0.8\},$$

where 1-cell is constructed via

$$|\rho_{ij,t}^{120}| \geq \tau_{\text{corr}}, \quad |\lambda_{i \to j,t}^{120,5}| \geq \tau_{\text{lead}}, \quad \phi_{ij,t} \geq \tau_{\text{beta}}.$$

If Rank IC, Sharpe ratio, transaction-cost-adjusted return (net return), cumulative P&L are sensitive in k or τ, this indicates that the 1-cell construction is unstable and should be reformulated.

1.5 Features and standarization

The 0-cochain is

$$x_{i,t} = \begin{pmatrix} r_{i,t-1} & \bar{r}_{i,t-6:t-2} & \bar{r}_{i,t-20:t-7} \\ \text{momentum}_{i,t}^{20} & \text{volatility}_{i,t}^{20} & \text{dollar\_volume}_{i,t}^{20} \\ \text{turnover}_{i,t}^{20} & \beta_{i,t-1}^{60} & \text{sector}_i \end{pmatrix},$$

where

$$\text{momentum}_{i,t}^{20} = \prod_{s=t-20}^{t-1} (1 + r_{i,s}) - 1,$$

$$\text{volatility}_{i,t}^{20} = \text{Std}(r_{i,t-20}, ..., r_{i,t-1}),$$

$$\text{turnover}_{i,t}^{20} = \frac{1}{20} \sum_{s=t-20}^{t-1} \frac{\text{dollar\_volume}_{i,s}}{\text{market\_capital}_{i,s}} = \frac{1}{20} \sum_{s=t-20}^{t-1} \frac{\text{volume}_{i,s}}{\text{share\_outstanding}_{i,s}},$$

$$\beta^{60}_{i,t} = \frac{\mathrm{cov}(\mathrm{return}_{i,t}, \mathrm{market\_return}_t)}{\mathrm{var}(\mathrm{market\_return}_t)}.$$

Let use $\cdot \mapsto z(\cdot)$ to denote cross-sectional z-score, and $\cdot \mapsto \omega(\cdot)$ to denote winsorize, where threshold is given by

$$\omega_f(x^{(f)}_{i,t}) = \min\Big\{\max\left(x^{(f)}_{i,t}, q_{0.01,f}\right), q_{0.99,f}\Big\},$$

where f is the placehold for feature.

return, momentum, beta $z(\omega(x))$

volatility $z(\omega(\log(x+\varepsilon)))$

dollar-volume $z(\omega(\log(1+x)))$

sector one-hot encoding $e_{s(i)}$, and learned embedding $E_{s(i),:}$.
Standarized 0-cochain

$$\tilde{x}_{i,t} = \begin{pmatrix} \tilde{r}_{i,t-1} \\ \tilde{\bar{r}}_{i,t-6:t-2} \\ \tilde{\bar{r}}_{i,t-20:t-7} \\ \widetilde{\mathrm{momentum}}^{20}_{i,t-1} \\ \widetilde{\mathrm{volatility}}^{20}_{i,t-1} \\ \widetilde{\mathrm{log\_dollar\_volume}}^{20}_{i,t-1} \\ \widetilde{\mathrm{log\_turnover}}^{20}_{i,t-1} \\ \tilde{\beta}^{60}_{i,t-1} \end{pmatrix} \oplus e_{s(i)} \oplus E_{s(i)}.$$

1.6 Temporal splitting

Rolling walk-forward split:

$$\text{train 8 years} \to \text{validate 1 year} \to \text{test 1 year},$$

then roll the window forward by one year.

2 Data-processing procedures

Step	Input	Output
Download	CRSP-API	raw data = (permno, date, prc, vol, ret, mkcap, OHLC, ..., mkret)
Canonicalize	raw data	$\rho_{i,t}$, (i,t) is the unique sorted permno-date key
Rank top500	$\rho_{i,t}$	$\mathcal{A}_t = \arg\max_{i_1 \geq \dots \geq i_{500}} \left\{ \overline{\text{dollar\_volume}}^{(21)}_{i,t} \right\}$
Feature-label	$\mathcal{A}_t$	$x_{i,t}, y_{i,t}$
Standardization	X_t	$\tilde{X}_t$
top-k, τ-threshold 1-cell	$\tilde{X}_t$	$K_t^{(1)} = \left\{ A_t^{\text{type}} \mid \text{type}=\cdots \right\}$
Chronological split	$K_t, \tilde{X}_t, y_{i,t}$	train-val-test folds $\mathcal{D}^{(k)}$
Ready-for-use	$\mathcal{D}^{(k)}$	$\left(\left(\tilde{X}_t, K_t \right), y_{i,t} \right)^{(k)}_{8,1,1}$

Table 2: Data-processing pipeline.

3 Models and Optimization Objective

3.1 Model filtration

The model filtration follows

$$\text{Logistic} \to \text{MLP} \to \text{GCN}.$$

The output of model has three layers:

$$z(x) = f_\theta(x) = (z_1, ..., z_5) \in \mathbb{R}^5,$$

$$\hat{p}(x) = \widetilde{\max}\left(z(x)\right) = \left(\frac{\exp(z_1)}{\exp(z_1) + \cdots + \exp(z_5)}, \cdots, \frac{\exp(z_5)}{\exp(z_1) + \cdots + \exp(z_5)} \right),$$

$$\hat{y} = \arg\max_k \hat{p}_k(x).$$

Logistic Model family

$$f_\theta(h) = \sigma(Wh + b),$$

where $\theta = (W, b)$.

MLP Model family

$$f_\theta(h) = \sigma \circ A^{(L)} \circ \cdots \circ \sigma \circ A^{(1)}(h),$$

where

$$A^{(l)}(h) = W^{(l)}h + b^{(l)}, \quad \theta = \left(W^{(l)}, b^{(l)} \mid 1 \leq l \leq L \right).$$

GCN A layer is

$$H^{(l+1)} = \sigma\left(\hat{A}H^{(l)}W^{(l)}\right),$$

where

$$\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}},$$
$$\tilde{A} = A + I,$$
$$\tilde{D} = \deg \tilde{A},$$
$$H^{(0)} = X_t = (x_{i,t})_{i\in\mathcal{A}_t},$$

and $W^{(l)}$'s are the parameters to be optimized.

3.2 Cross_entropy optimization objective

The pointwise categorical cross-entropy loss is

$$\ell(\hat{y}_{i,t}, y_{i,t}) = -\log\left(\hat{p}_{i,t}^{(y_{i,t})}\right).$$

The empirical risk is

$$\hat{R}(\theta) = \frac{1}{T}\sum_{t\in\mathcal{T}}\frac{1}{|A_t|}\sum_{i\in\mathcal{A}_t}\ell_{i,t}(\theta).$$

The cost may contain a penalty term

$$J(\theta) = \hat{R}(\theta) + \lambda\Omega(\theta).$$

4 Portfolio

4.1 Ranking score

For ranking and portfolio construction, predicted class probabilities are converted into a scalar score:

$$\hat{s}_{i,t} = \sum_{c=-2}^{2} c\hat{p}_{i,t}^{(c)}.$$

This score is the model_implied expected class value. Classification loss trains the model, while the continuous score is used for Rank IC, Pearson IC, and portfolio ranking. This makes the pipeline consistent from classification to trading evaluation.

4.2 Long-short portfolio

At each t, assets are ranked by the scalar score $\hat{s}_{i,t}$. Let L_t be the top 20-quantile of $\mathcal{A}_t$, and S_t be the bottom 20-quantile assets. The portfolio is equal-weighted, dollar-neutral, and rebalanced daily:

$$w_{i,t} = \begin{cases} \frac{1}{2|L_t|}, & i \in L_t \\ \frac{1}{-2|S_t|}, & i \in S_t \\ 0, & \text{o.w.} \end{cases}.$$

The 1-horizon portfolio return is

$$R_{t+1} = \sum_{i \in \mathcal{A}_t} w_{i,t} r_{i,t+1}.$$

The net return is

$$\pi_{t+1} = R_{t+1} - \kappa \sum_{i \in \mathcal{A}_t \cup \mathcal{A}_{t-1}} |w_{i,t} - w_{i,t-1}|,$$

where κ is the one-way transaction cost per dollar traded.

5 Report

5.1 Ablation

To evaluate whether additional model structure improves out-of-sample alpha prediction, the experiment uses the ablation table below. Each model keeps the same prediction target and chronological evaluation protocol, but changes the representation layer or cross-asset structure available to the model.

Model	**Input**	**1-cell Structure**
M0 Logistic	0-cochain	None
M0-Plus XGboost	0-cochain	None
M1 MLP	0-cochain	None
M2-A GCN-Corr	0-cochain	Correlation
M2-B GCN-LeadLag	0-cochain	Lead-lag
M2-C GCN-Sector	0-cochain	Sector
M2-D GCN-SupplyChain	0-cochain	Supply-chain
M2-E GCN-Beta	0-cochain	Market exposure
M2-0 GCN-Random	0-cochain	Random control
M2-Pro Multi-Rel GCN	0,1-cochain	Multi-relation
M3 Sheaf GCN	0,1-cochain, restriction map	Multi-relation consistency

Table 2: Ablation design for the model filtration.

5.2 Evaluation

Metrics are reported separately on validation and test periods for each model in the ablation table. The primary metrics are Rank IC, Sharpe ratio, net return, and cumulative P&L. The secondary metrics are Pearson IC, classification accuracy, turnover, and maximum drawdown.

Primary metrics Rank IC is

$$\mathrm{RankIC}_t = \mathrm{SpearmanCorr}\left(\{\hat{s}_{i,t}\}_{i \in \mathcal{A}_t}, \{r_{i,t+1}\}_{i \in \mathcal{A}_t}\right).$$

The Rank IC reported in the tables is its time_series average,

$$\overline{\mathrm{RankIC}} = \frac{1}{T} \sum_{t \in \mathcal{T}} \mathrm{RankIC}_t\,.$$

Rank IC is the main prediction metric because it evaluates whether the score correctly orders assets in the 1_horizon cross_asset, without requiring a linear relationship between the scores and realized returns.
The annualized Sharpe ratio is

$$\text{Sharpe} = \sqrt{252}\,\frac{\text{mean}(\pi)}{\text{std}(\pi)}.$$

The net return reported in the tables is the mean daily net return

$$\bar{\pi} = \frac{1}{T}\sum_{t\in\mathcal{T}} \pi_{t+1},$$

expressed in basis points per day.
The cumulative P&L at the end of the evaluation period is

$$\text{P\&L} = \prod_{t\in\mathcal{T}} (1 + \pi_{t+1}) - 1.$$

Secondary metrics Pearson IC is

$$\text{IC}_t = \text{Corr}\left(\{\hat{s}_{i,t}\}_{i\in\mathcal{A}_t}, \{r_{i,t+1}\}_{i\in\mathcal{A}_t}\right).$$

The reported Pearson IC is its time_series average,

$$\overline{\text{IC}}\frac{1}{T}\sum_{t\in\mathcal{T}} \text{IC}_t\,.$$

Multiclass classification accuracy is

$$\text{Acc}_t = \frac{1}{|\mathcal{A}_t|}\sum_{i\in\mathcal{A}_t} \mathbb{1}\{\hat{y}_{i,t} = y_{i,t}\},$$

and the reported accuracy is

$$\text{Acc} = \frac{1}{T}\sum_{t\in\mathcal{T}} \text{Acc}_t\,.$$

Turnover is

$$\text{TO}_t = \sum_{i\in\mathcal{A}_t\cup A_{t-1}} |w_{i,t} - w_{i,t-1}|,$$

and reported version is the mean daily turnover.
To define maximum drawdown, let

$$M_t = \max_{u\leq t} V_u,$$

where

$$V_u = \prod_{u'\leq u} (1 + \pi_{u'}).$$

Define the drawdown at t is

$$D_t = 1 - \frac{1 + V_t}{1 + M_t},$$

and the maximum drawdown over the evaluation period is

$$\text{MDD} = \max_t D_t = \max_t \left(1 - \frac{1 + V_t}{1 + \max_{u\leq t}} V_u\right).$$

6 Execution

1. Follow data-processing procedures.
2. Train, validate, and retrain models on the same rolling walk-forward splits.
3. Convert probabilities to ranking scores and daily long-short portfolios.
4. Evaluate Rank-IC, Sharpe, net-returns, P&L, IC, turnover, drawdown, and accuracy on test dataset.
5. Evaluate the sensitivity by variation of the metrics with differet k and τ.
6. Run ablations, then report whether gains justify added complexity.